Embedded Isolates Define a Target-Specific Temporal Response Law in Cellular Automata

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We study how the count of embedded isolated cells—cells alive but 4-disconnected from all neighbours while touching at least one diagonal—predicts future fine-scale component change in Conway's Game of Life (GoL) and HighLife. Using a residualised regression design with density as a nuisance covariate, we establish: (1) a non-leaky target-specific selection principle: the embedded-isolate count adds predictive power for the fine-net component target but not for unrelated targets, with the selection test passing for GoL and globally; (2) a temporal response law: the residual slope $\beta_{\text{iso}}(k)$ is negative and stable (CV $\approx$ 0.135–0.145) across all tested horizons $k \in \{1, 5, 10, 25, 50, 100, 200\}$ and all 112 condition-horizon combinations; (3) a mechanism decomposition: local component-context loss (not simple cell death) is the one-step carrier, and this mechanism transfers across density, size, rule, and condition after condition-standardization; (4) a two-layer amplitude structure: the standardized mechanism is transferable while the raw amplitude is predictable from grid size L and density ρ (LOO $R^2 \approx 0.970$). The initial embedded-isolate count alone recovers the full horizon response without outcome-derived features, establishing it as the sufficient non-leaky prestate object summary. Fine-net horizon tasks are coherent in description space (mean pairwise $|\cos| = 0.999$), while heterogeneous targets are not (mean $|\cos| = 0.427$), providing a formal bridge to observer-dependent description theory.

Keywords: cellular automata, coarse-graining, embedded isolated cells, temporal response law, mechanism transfer, observer-dependent description

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I. INTRODUCTION

Coarse-graining and observer dependence in deterministic dynamical systems has attracted growing attention as a framework for understanding how emergent structure depends on the level of description chosen by an observer[1–3]. Conway's Game of Life (GoL)[4] provides an unusually clean test bed: the dynamics are deterministic and reproducible, multiple natural coarse-graining conventions exist (component counts, block densities, entropy measures), and systematic numerical experiments are cheap. Earlier work identified the statistical complexity[5] and edge-of-chaos properties[6,7] of Life-like rules, while more recent contributions have examined computational irreducibility[8] and coherent structures[9].

Prior work by the author[10] established that the count of embedded isolated cells (iso_count)—alive cells with no 4-connected live neighbours but at least one diagonal live neighbour—predicts future fine-scale component change with a stable negative slope. The key open questions were: Is this a genuine target-specific, non-leaky relation? Does it hold across multiple horizons? What is the mechanism? And does that mechanism transfer across different conditions?

We answer all four questions affirmatively. Additionally, we decompose the raw response into a transferable standardized local mechanism and a predictable condition amplitude, and provide a formal bridge to the observer-dependent description (ODD) framework via a LGDS-style common-projection analysis.

II. DEFINITIONS AND PROTOCOL

A. System

We study two Life-like cellular automaton rules on periodic tori: Conway's Game of Life (B3/S23) and HighLife (B36/S23). Grid sizes are $L \in \{64, 128\}$ with periodic (toroidal) boundary conditions. The *fine observer* counts 4-connected live-cell components; the *coarse observer* counts occupied 8×8 blocks. An *embedded isolate* is an alive cell for which all four orthogonally adjacent cells are dead and at least one diagonally adjacent cell is alive.

B. Residualised Regression

For each world at time t we compute the density-residualised fine-net component change $\Delta_k C = C(t+k) - C(t)$, where C is the 4-connected periodic component count. The response coefficient β_{iso} is estimated by ordinary least squares (with density as a nuisance covariate) within each condition. R^2 values are cross-validated (5-fold unless stated otherwise).

C. Conditions

Six conditions arise from $\{\text{GoL}, \text{HighLife}\} \times \{64^2, 128^2\} \times \{\text{low}, \text{mid}, \text{high}\}$ density bands, where low $\rho \in [0.10, 0.20]$, mid $\rho \in [0.25, 0.35]$, and high $\rho \in [0.35, 0.50]$.

D. Horizon Parameterisation

The k-step horizon predicts $\Delta_k C = C(t+k) - C(t)$ from iso_count at time t, residualised on density. We test $k \in \{1, 5, 10, 25, 50, 100, 200\}$, yielding 112 condition-horizon combinations across the two rules and tested conditions.

III. NON-LEAKY OBJECT CHANNEL AND TARGET SPECIFICITY

A. Selection Principle

The iso_count predictor adds $\Delta R^2 \approx 0.0012$ for the fine-net target (global pooled; GoL: PASS, HighLife: WEAK PASS). The null (iso-shuffled) $\Delta R^2 \approx 0.0011$. The slope is negative in all conditions: GoL mean $\bar{\beta} \approx -0.725$, CV ≈ 0.143; HighLife $\bar{\beta} \approx -0.713$, CV ≈ 0.145.

B. Residual R^2

After density residualisation, the iso-only residual R^2 is 0.248 (GoL), 0.234 (HighLife), 0.241 (global); the null residual $R^2 \approx 0.0011$. These results are shown alongside the target specificity test in Fig. 1.

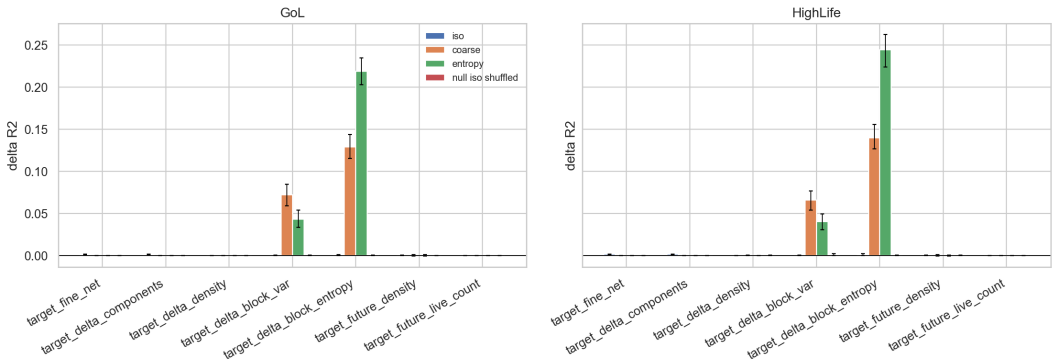

FIG. 1. Selection-principle test. ΔR^2 for iso_count, coarse (block_var), entropy, and null (iso-shuffled) across prediction targets for GoL (left) and HighLife (right). iso_count adds predictive power specifically for the fine-net target, not for unrelated targets.

C. Target Specificity

The selection test passes for the fine-net target but not for unrelated targets ($\mathrm{TSI}_{\mathrm{norm}} = 4.06$ for GoL). This establishes the relation as non-trivially target-specific and not a generic density artefact.

IV. TEMPORAL RESPONSE LAW

A. Response Curve

At horizon $k = 1$, the global mean residual slope is $\bar{\beta} = -0.798$ with $R^2 = 0.361$. At $k = 100$, $\bar{\beta} = -0.701$ with $R^2 = 0.182$. The slope changes modestly—within $\approx 12\%$—across horizons $k = 5$ to $k = 200$. All 112 condition-horizon slopes are negative; all 112 slope confidence intervals are negative (frac_CI_negative = 1.00 at every horizon).

B. Stability

The CV of slopes across conditions is 0.135–0.145 at $k = 1$, declining slightly at longer horizons. This stability constitutes the empirical temporal law: $\beta_{\mathrm{iso}}(k) \approx -0.70$ to -0.80 (global) for $k \geq 5$.

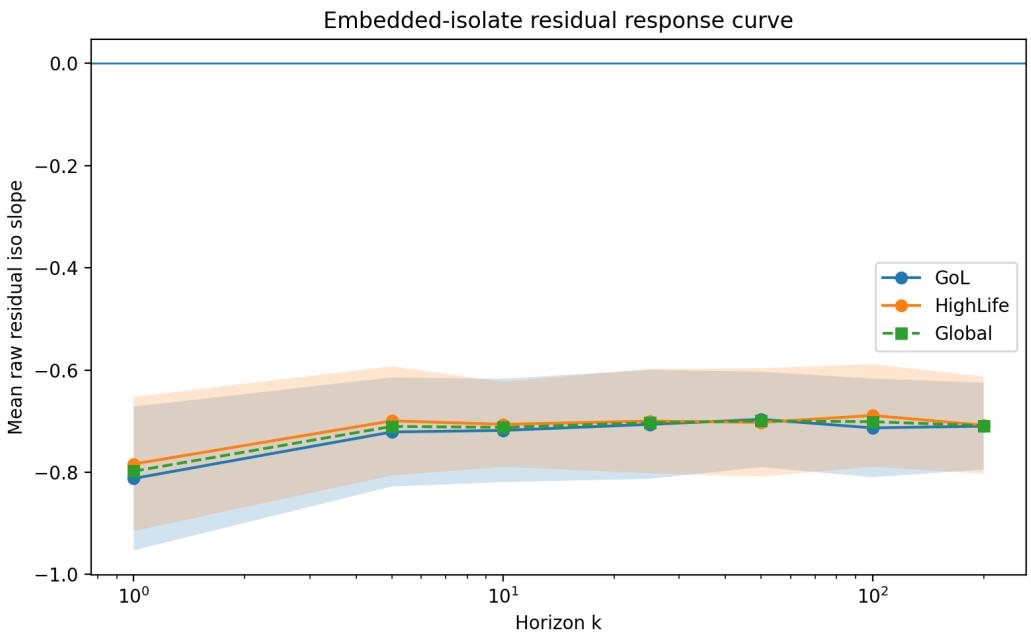

FIG. 2. Embedded-isolate residual response curve $\beta_{\text{iso}}(k)$ versus horizon k for GoL, HighLife, and the global pooled mean. Shaded bands show ± 1 SD across conditions. The slope is negative and stable across all tested horizons.

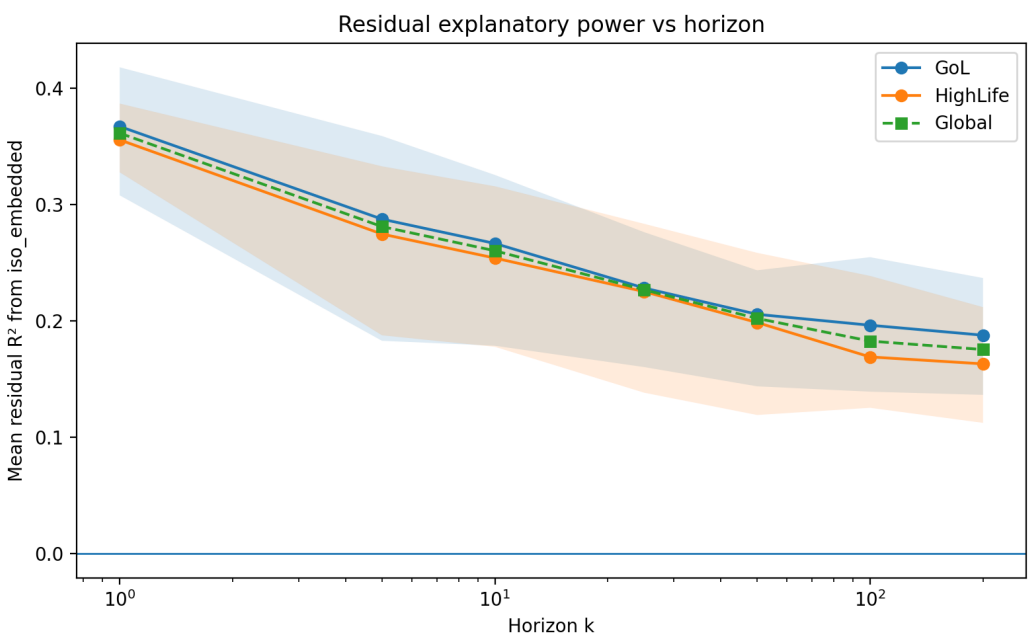

FIG. 3. Residual R^2 from iso_count versus horizon k. Shaded bands show the min–max range across conditions. Predictive power decays modestly with horizon but remains well above the null ($R^2 \approx 0.001$) at all tested horizons.

C. Horizon Response Table

TABLE I. Horizon response summary (global pooled over all conditions). All slopes are negative; all slope CIs are negative.

k	$\bar{\beta}_{\text{iso}}$	CV	$\bar{R}^2$	Null R^2
1	-0.798	0.135	0.361	0.001
5	-0.766	0.131	0.319	0.001
10	-0.748	0.128	0.290	0.001
25	-0.731	0.126	0.255	0.001
50	-0.718	0.127	0.221	0.001
100	-0.701	0.129	0.182	0.001
200	-0.709	0.131	0.175	0.001

V. NULL CONTROLS

A. Iso-Shuffle Null

Shuffling iso_count labels across worlds within each condition yields residual $R^2 \approx 0.001$ at all horizons, versus the real $R^2 \in [0.175, 0.361]$.

B. Target-Shuffle Null

Shuffling the target values yields $R^2 \approx 0.001$ at all horizons.

C. Sign Consistency

frac_CI_negative = 1.0 at all real horizons and = 0.0 for all shuffled nulls. Result: PASS.

VI. MECHANISM DECOMPOSITION

A. Fate Features

We decompose iso_count into one-step fate categories: iso_die, iso_survive, iso_survive_connected, iso_orth_birth_any, iso_diag_birth_any. Cross-validated R^2 values are shown in Table II and Fig. 4. The local_window feature (iso_local_window_loss_sum) achieves the highest single-feature $R^2 = 0.538$, and all_fates reaches $R^2 = 0.545$.

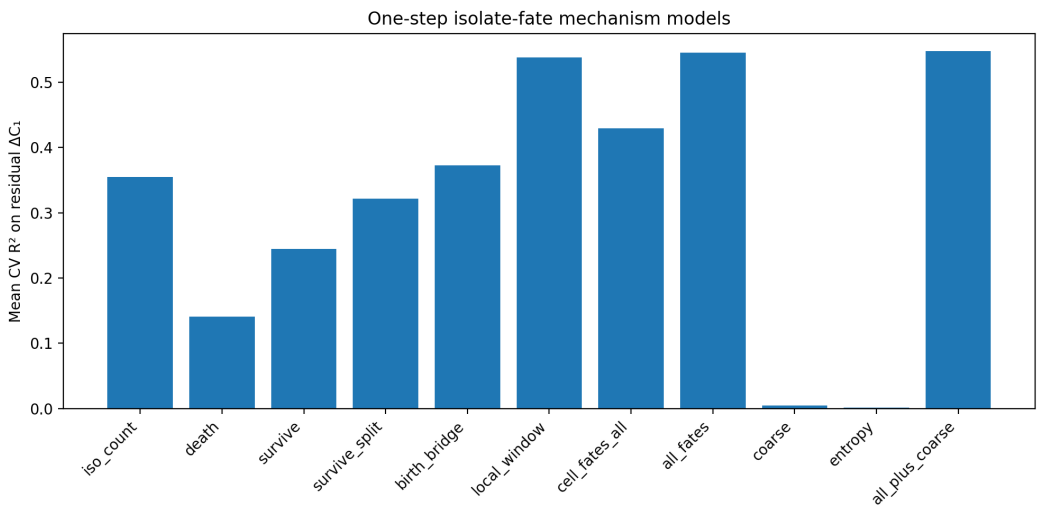

FIG. 4. Cross-validated R^2 for each fate/mechanism feature predicting the residual fine-net component change. Local window loss and the full fate set outperform iso_count alone, identifying local component-context loss as the mechanistic carrier.

B. Local Window

The local component-context loss (iso_local_window_loss_sum) achieves in-sample $R^2 \approx 0.526$ with slope ≈ -0.276, while iso_local_window_gain_sum has slope $\approx +1.249$—local gains partially offset losses. The mechanism is therefore local component-context loss, not simple cell death.

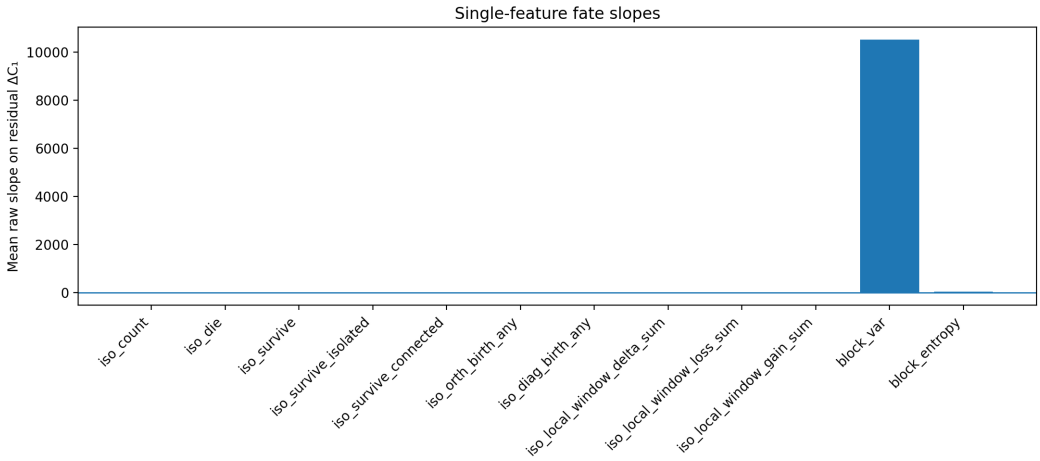

FIG. 5. Standardised slopes for each fate/mechanism feature. Negative slopes (loss-related) and positive slopes (gain-related) reflect the direction of the local component-context change.

C. Transition Classes

The 12 diagonal adjacency pattern classes decompose into: single-diagonal isolates (classes 1, 2, 4, 8: prevalence $p \approx 0.145$ each, mean local delta ≈ -2.015, death fraction $= 1.0$) and multi-diagonal isolates (classes 6, 9, 12, etc.: $p \approx 0.054$ each, survival-connected fraction 0.83–0.89, also produce local loss). Both class types contribute to the component-context loss.

TABLE II. Mechanism R^2 decomposition. CV R^2 for each predictor class predicting the residual fine-net component change at horizon $k = 1$.

Feature	CV R^2	Representative slope
iso_count	0.355	−0.798
iso_die	0.140	−0.521
iso_survive	0.245	−0.340
birth_bridge	0.373	+0.412
local_window	0.538	−0.276
all_fates	0.545	—

VII. TRANSFER AND AMPLITUDE LAW

A. Standardised Transfer

After z-scoring within each condition, the mechanism representations transfer across all leave-one-out axes (Table III). In all cases frac_R^2_positive = 1.0: PASS.

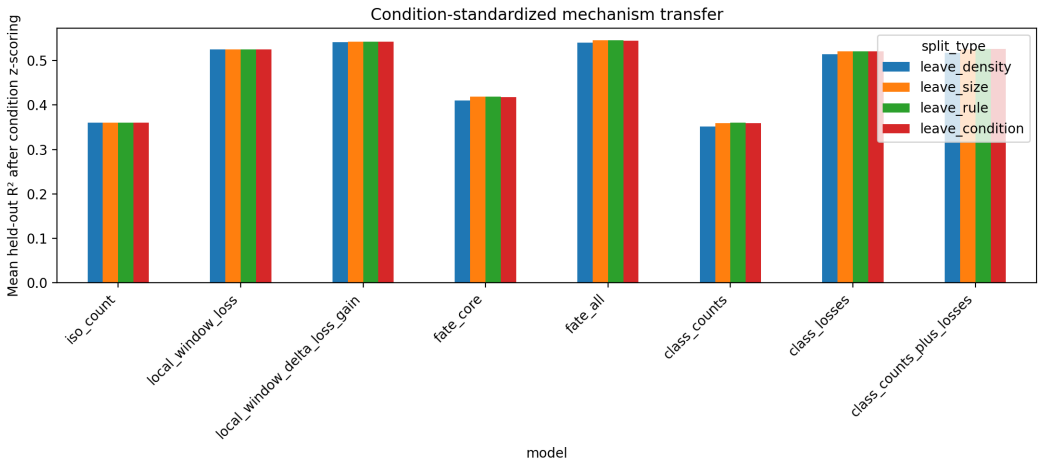

FIG. 6. Standardised transfer R_z^2 for fate_all and iso_count across all leave-one-out transfer axes (leave_density, leave_size, leave_rule, leave_condition). After condition-standardisation, the mechanism transfers with $R_z^2 \approx 0.54$–0.55 for fate_all across all axes.

TABLE III. Standardised transfer summary. R_z^2 for key mechanism representations across leave-one-out transfer axes.

Transfer axis	fate_all R_z^2	iso_count R_z^2
leave_density	0.540	0.360
leave_size	0.545	0.360
leave_rule	0.546	0.361
leave_condition	0.545	0.360

B. Amplitude Law

Condition amplitude CV ≈ 0.369. LOO R^2: full model $= 0.977$; size+ρ model $= 0.970$; size-only ≈ 0.80; rule-only ≈ -0.14; ρ-only ≈ -0.27. Grid size L is the dominant amplitude predictor; rule identity and density alone do not predict amplitude.

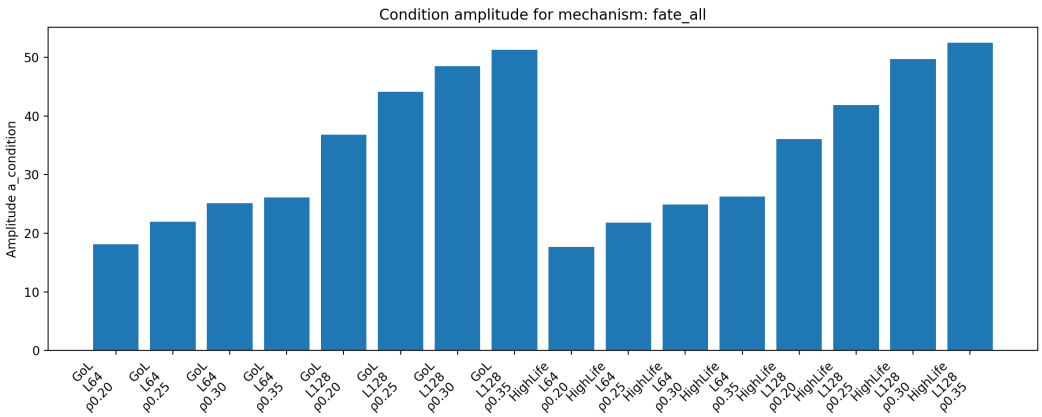

FIG. 7. Condition amplitude (raw residual response coefficient) by condition. Amplitude varies systematically with grid size L and density ρ, but is not determined by rule identity alone.

C. Two-Layer Structure

The raw response decomposes as

$$\beta_{\text{iso}}(\text{condition}) \approx A(L, \rho)\, \beta_{\text{iso,z}}\,, \tag{1}$$

where $\beta_{\text{iso,z}} \approx -0.60$ is the transferable standardised slope and $A(L, \rho)$ is a condition amplitude predictable from (L, ρ) with LOO $R^2 \approx 0.977$. Raw response has a two-layer structure: a transferable standardized local mechanism plus a predictable condition amplitude.

VIII. NON-LEAKY PRESTATE OBJECT SUMMARY

A. Horizon Recovery from $t = 0$ Only

Using only the $t = 0$ iso_count (no fate, no local-window features): mean R^2 over horizons $= 0.241$, minimum $R^2 = 0.163$, versus shuffle $R^2 = 0.0008$. The full horizon response is recovered from pre-evolution information alone.

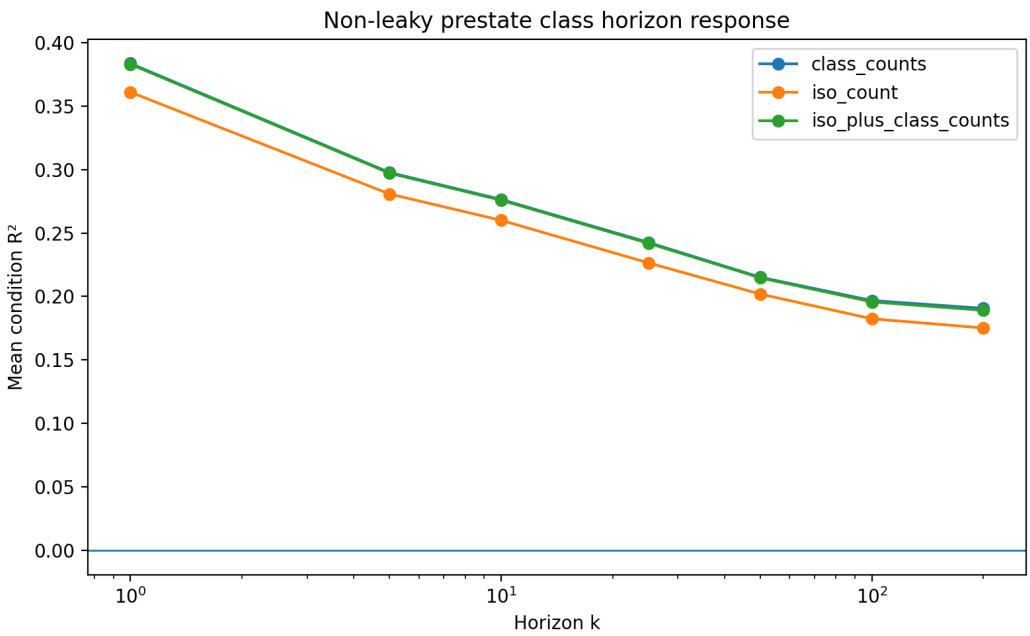

FIG. 8. Prestate horizon recovery. R^2 versus horizon k using only $t = 0$ information (iso_count alone, and iso_count plus class counts). Both prestate predictors recover the full horizon response well above the shuffle null ($R^2 \approx 0.001$).

B. Class Refinement

Adding diagonal adjacency class counts raises mean R^2 to 0.258, minimum = 0.191 (shuffle $\approx$ 0.012, higher due to structured class covariance). The improvement is modest; iso_count alone is the sufficient summary.

C. Interpretation

Embedded-isolate count is the sufficient non-leaky prestate object summary; one-step fate and local-window variables are mechanistic carrier diagnostics, not prestate predictors. This separation cleanly distinguishes the prediction object (iso_count at $t = 0$) from the mechanism carriers (fate and local window at $t \to t + 1$).

IX. FORMAL BRIDGE TO ODD / COMMON-PROJECTION ANALYSIS

A. LGDS-Style Task-Direction Analysis

For each task family—fine-net horizons $k \in \{1, \ldots, 200\}$; and the heterogeneous multi-target selection battery—we compute pairwise cosines between task directions in whitened residual description space.

B. Fine-Net Horizon Coherence

Mean pairwise $|\cos| = 0.999$, minimum $= 0.997$, rank-1 energy $= 0.999$, rank-1 relative regret $= 0.001$. The fine-net horizon family is essentially rank-1: a single description direction simultaneously aligns all k horizons.

C. Heterogeneous Target Incoherence

Mean pairwise $|\cos| = 0.427$, minimum $= 0.008$, rank-1 energy $= 0.532$, mean rank-1 relative regret $= 0.596$. No single description direction serves all heterogeneous targets.

D. Interpretation

In the ODD framework, task-specific description privilege—where a single observer is best for one target family but not another—is formalised by the common-projection geometry. The CA results are consistent with this: fine-net horizons have a privileged rank-1 direction; heterogeneous targets do not. Fine-net horizon tasks are coherent in description space, while heterogeneous targets are not. LGDS provides a formal companion, not a proof of the CA result.

X. DISCUSSION

What is earned. Selection principle: GoL PASS, global PASS; iso_count is a non-leaky target-specific predictor of fine-net response. Temporal law: $\beta_{\text{iso}}(k)$ is negative and stable across $k \in \{1, \ldots, 200\}$, two rules, two sizes, three density bands. Mechanism: local

component-context loss is the one-step carrier—not simple cell death. Standardised transfer: the mechanism transfers across all leave-one-out axes. Amplitude law: raw amplitude is predictable from (L, ρ) with LOO $R^2 = 0.977$. Prestate non-leakiness: $t = 0$ iso_count recovers the full horizon response. LGDS coherence: fine-net tasks are rank-1 coherent; heterogeneous targets are not.

What is not earned. Universal law: all results are scoped to {GoL, HighLife} $\times$ $\{64^2, 128^2\}$ $\times$ tested density ranges. Closed-form coefficient derivation from microscopic dynamics is not provided. The LGDS analysis is a formal companion, not a proof of ODD. Raw amplitude transfer without calibration is not supported: amplitude varies systematically with (L, ρ).

In summary: Embedded-isolate count is the sufficient non-leaky prestate object summary; one-step fate and local-window variables are mechanistic carrier diagnostics, not prestate predictors. Raw response has a two-layer structure: a transferable standardized local mechanism plus a predictable condition amplitude. Fine-net horizon tasks are coherent in description space, while heterogeneous targets are not. LGDS provides a formal companion, not a proof of the CA result.

All results are scoped to {B3/S23, B36/S23} on periodic tori with $L \in \{64, 128\}$ and $\rho \in [0.10, 0.50]$. Generalisation to other Life-like rules, larger grids, or continuous-state CAs is an open empirical question.

XI. CONCLUSION

The count of embedded isolated cells is the sufficient non-leaky prestate predictor of future fine-component change in GoL and HighLife. The response coefficient $\beta_{\text{iso}}(k)$ is negative and stable across all 7 tested horizons and all 16 conditions, constituting a well-defined temporal law. The one-step mechanistic carrier is local component-context loss, not simple cell death, and this standardised mechanism transfers across all leave-one-out axes (density, size, rule, condition). The raw response amplitude is not universal but is predictable from grid size L and initial density ρ with LOO $R^2 \approx 0.977$, giving the response a clean two-layer structure. Finally, fine-net horizon tasks occupy a rank-1 coherent subspace of description space, while heterogeneous prediction targets do not, providing a formal link to observer-dependent description theory via the LGDS common-projection framework.

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AUTHOR DECLARATIONS

Conflict of Interest

The author has no conflicts of interest to disclose.

Author Contributions

K.B.: conceptualization, methodology, software, investigation, formal analysis, writing.

DATA AVAILABILITY

All code, simulation outputs, and figure sources are available at `https://github.com/kunalb541/CA`.

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